ELLIOT HAWKINS

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EDUCATION

Master of Science in Electrical Engineering and Computer Science; Electrical Engineering Emphasis **May 2027**
Fowler School of Engineering, Chapman University, Orange, CA
Bachelor of Science in Electrical Engineering; Minor: Music Technology **May 2026**
Fowler School of Engineering, Chapman University, Orange, CA 3.92 GPA

RELEVANT COURSES

Electronics and Circuits II | Control Systems | Electromagnetics I | Digital Logic Design II | Systems Programming
Engineering Mathematics | Microelectronics | Robotics | Digital Signals and Filters | Linear Algebra and Differential Equations

TECHNICAL SKILLS

Circuit Analysis | PCB Design and Fabrication | Python and C/C++ | Embedded Systems | Soldering | Laser Cutting | 3D Printing
Signal Processing | Arduino | 3D CAD | Project Management | MatLab | System Verilog | Waveform Analysis | Adobe Suite | CNC

EXPERIENCE

Student Makerspace Lead Employee, (DCI Lab) | Chapman University **September 2022 - (Lead)August 2025 - Present**
- Guided and supported student projects by providing expertise in:
  - PCB | CNC and milling machines | Circuit analysis and testing | 3D printing and resin printing | Laser cutting | Soldering | Communication and organization | Poster and sticker printing | Wood and metal fabrication | Post-production | Project Management
- Developed and maintained comprehensive, streamlined instructions for equipment operation and sample project implementation
- Developed and directed hands-on interactive workshops in:
  - Laser Cutting (File set-up and process) | Soldering (Introduction to PCB soldering) | Introduction to 3D printing (printing process) | Introduction to Resin Printing (printing process)
  - Project workshops:
    - "Useless Machine" (Soldering | electronics | laser cutting skills) | Anti-Gravity figure (3D modeling | printing)

Teacher's Assistant, Electromagnetics II (EENG-430) | Chapman University **Fall 2024**
- Modeled antennas using Ansys HFSS software for advanced electromagnetic field simulations
- Developed curriculum support materials to assist the professor in enhancing student learning
- Conducted office hours to provide personalized software assistance to students

Teacher's Assistant, Introduction to PCB design and Fabrication (CPSC-298) | Chapman University **Fall 2025**
- Guided students through PCB design and fabrication on LPKF tools: ProtoMat and ProtoLaser

PROJECTS

Target Retrieval System for Gun Ranges (Independent Contracting) **December 2025-Present**
- Designed industrial motor control system with touchscreen HMI, PLC integration, and three-phase motor command output
- Programmed industrial PLCs for real-time motor control logic including speed and operational sequencing
- Designed custom PCB for signal conditioning and voltage regulation to interface with motor controller with high efficiency
- Design a custom User Interface on the HMI utilizing TTL UART RX/TX to Modbus 485 serial communication conversion
- Configured multi-voltage power distribution from 110V AC mains to regulated DC rails, integrating circuit breakers and existing power supply hardware for safe system operation

Remote Controlled Life-Sized Functional R2-D2 Replica from Star Wars (Robotics Club) **February-August 2025**
- Project lead and head of operations in all sub-groups
- Designed and fabricated custom PCBs for R2-D2's lighting systems with custom sequences on programmable integrated circuits
- Collaborated with teammates to fabricate a control system that integrated all R2-D2 functionality, including movement and sound
- Utilized Raspberry Pi's networking capabilities to fabricate a custom controller equipped with a user interface
- Gained problem-solving experience in every aspect of the project, including post-processing, electrical, and software areas

"**Useless Machine" Workshop** (DCI workshop) **January - May 2024**
- Developed a machine to contact and engage a switch with a motor-controlled plank when initiated by the user
- Utilized electronic components that operated without requiring programming, microcontrollers, or Arduino-based systems
- Expanded knowledge of electronic housing, fabrication, engineering process with project management, and circuit development
- Streamlined the manufacturing process and facilitated workshops for students to assemble their own

ACHIEVEMENTS & ACTIVITIES

Chapman Robotics Club, **Secretary** Fall 2024 - Present
Society of Manufacturing Engineers, **Secretary** Spring 2025 - Present
Fowler School of Engineering Deans Scholarship, **Recipient** Fall 2022 - Present
Chapman University Provost List, **Member** Spring 2023, Spring 2024, Fall 2024
Chapman University Computer Science Club, **Member** Fall 2022 - Present
Chapman University Panther Game Development, **Member** Fall 2024 - Present
World Cube Association, **Staff Member, and Competitor** 2017 - Present